

The background features a dark blue area on the left with vertical green wavy lines. On the right, there is a light gray area. A diagonal line separates these two areas, with several overlapping translucent bands in shades of gray and teal running along it.

Atomic Energy Levels

Key Objectives

- The Bohr model of the hydrogen atom integrates light quanta and atomic energy states to explain the specific wavelengths in the hydrogen spectrum and in the spectra of other simple atoms; the Bohr model enables line spectra to be correlated with atomic energy-level diagrams

Relevant Equations

$$\Delta E = hf$$

$$E_2 - E_1 = hf$$

Text References

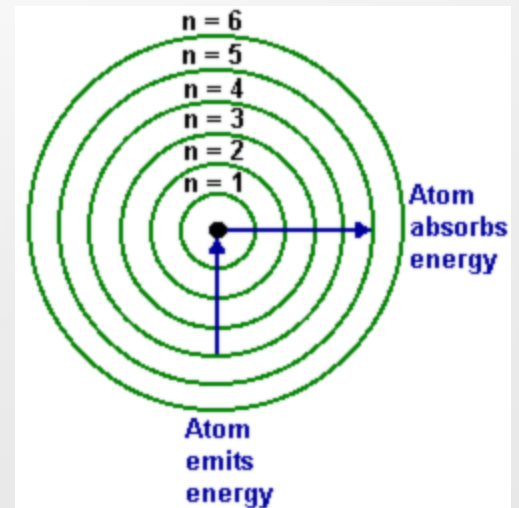
- STAWA Set 13

Bohr's model of the atom

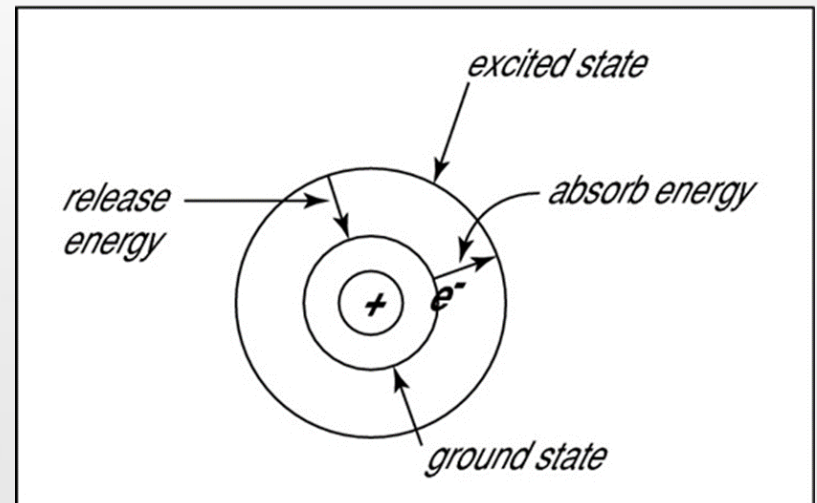
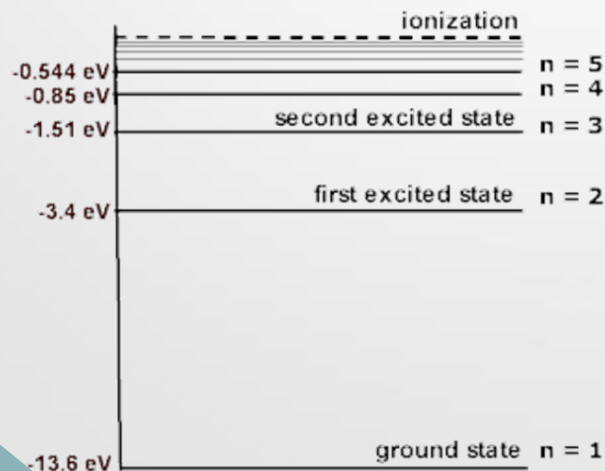
- In Bohr's model electrons can only orbit in stable shells (energy levels).
- Electrons can move between the shells but they require energy to reach higher energy levels.
- When moving in stable orbits electrons do not radiate energy. Energy is only radiated/absorbed when an electron jumps between the stable orbits.

$$E_2 - E_1 = hf$$

- In physics we use energy level diagrams to show the energy difference between the shells.

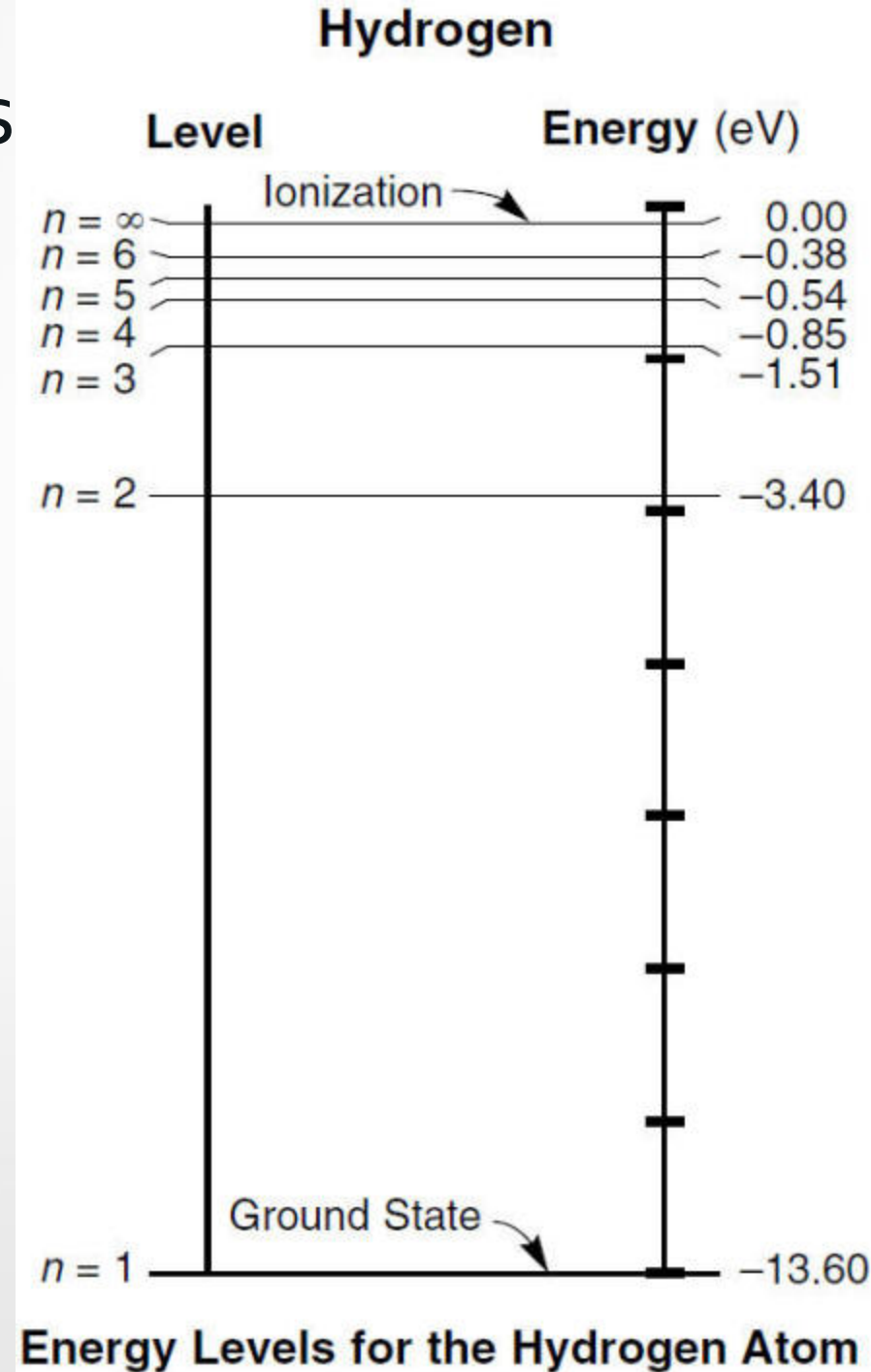


- $n=1$ is the closest shell to the nucleus and is also called the ground state.
- For hydrogen the energy level for $n = 1$ is -13.6 eV .
- If an electron moves up to a higher level (eg from $n = 1$ to $n = 2$) then it said to be in an excited state.
- As an electron moves to higher energy levels the energy becomes less negative until it reaches zero (ionisation).
- This is similar to gravitational potential energy, as electrons fall down towards the nucleus they lose potential energy, in order to move up to a higher energy level they must gain potential energy.



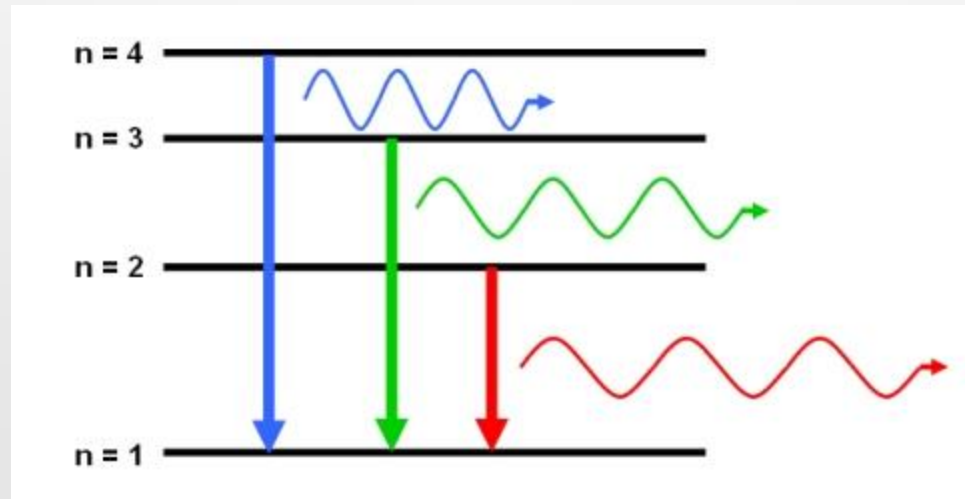
Hydrogen energy levels

- An electron in the ground state requires 13.6 eV of energy to move to $n = \infty$.
- This is the ionisation energy, the energy required to release an electron from the hydrogen atom.
- Draw this diagram so you can use it for calculations throughout this PowerPoint.
- You do not need to memorise this diagram, it is given in questions of the exam.

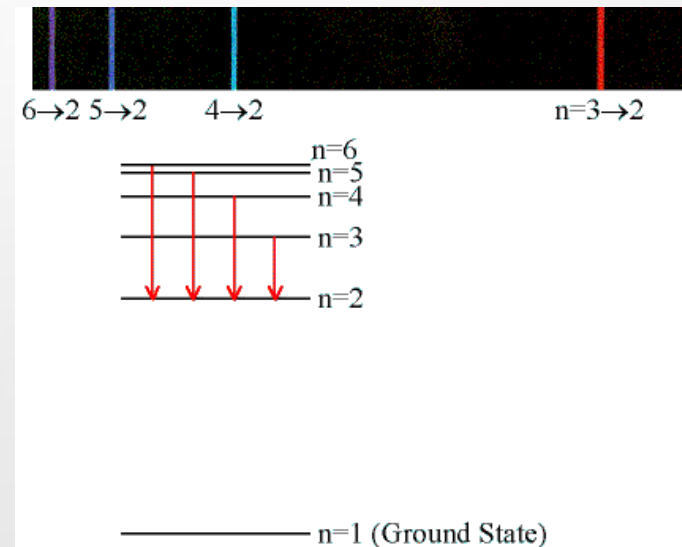
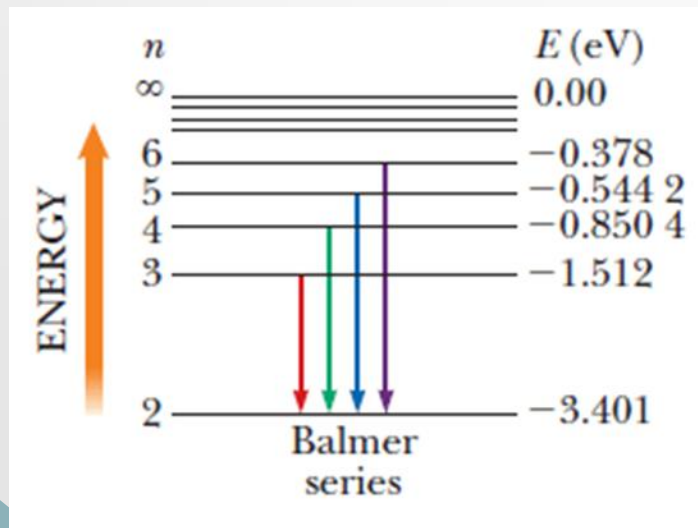


Emission

- When an electron falls to a lower energy level it emits a photon.
- The energy lost by the electron must be released as the energy of the photon (conservation of energy).
- We can calculate the frequency, and therefore determine the colour of emitted photons using $\Delta E = hf$



- The emission of photons explains the line emission spectra for hydrogen.
- An element can produce emission lines of specific colours that correspond to the allowed energy transitions for that atom. These transitions are unique to that element hence the unique atomic emission spectra observed for each element.
- The transition to a lower energy level from $n=6$ to $n=2$ corresponds to visible lines in hydrogens emission spectra.

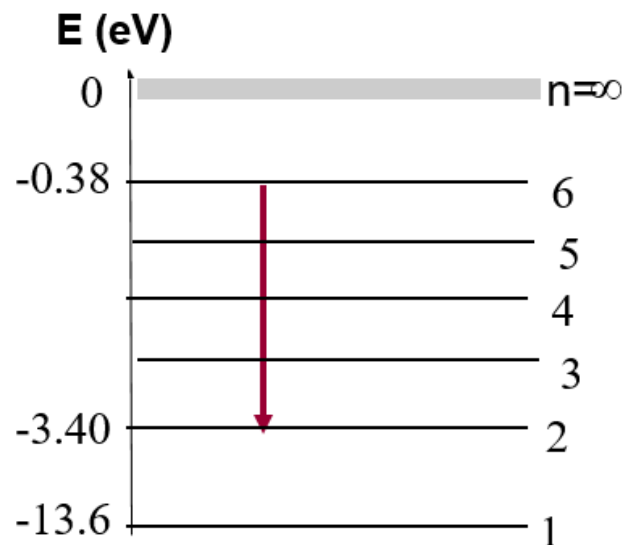


- Hydrogen emits a violet photon for the transition from the $n = 6$ to the $n = 2$ energy level.
- For this transition the atom loses energy of:

$$\Delta E = 3.4 - 0.38 = 3.02 \text{ eV} = 4.83 \times 10^{-19} \text{ J}$$

$$\Delta E = hf \text{ so } f = \frac{\Delta E}{h} = \frac{4.83 \times 10^{-19}}{6.63 \times 10^{-34}} = 7.28 \times 10^{14} \text{ Hz}$$

This can also be written as 728 THz, which is a violet frequency of visible light.



The colors of the visible light spectrum^[1]

color	Wavelength interval	Frequency interval
Red	~ 700–635 nm	~ 430–480 THz
Orange	~ 635–590 nm	~ 480–510 THz
Yellow	~ 590–560 nm	~ 510–540 THz
Green	~ 560–520 nm	~ 540–580 THz
Cyan	~ 520–490 nm	~ 580–610 THz
Blue	~ 490–450 nm	~ 610–670 THz
Violet	~ 450–400 nm	~ 670–750 THz

- Calculate the wavelength emitted when an electron makes the following transitions and determine the colour of the emitted photon.
 - $n=3$ to $n=2$ 658 nm red
 - $n=4$ to $n=2$ 488 nm blue
 - $n=5$ to $n=2$ 435 nm violet
- Why can we not see a photon corresponding to an energy transition from $n = 2$ to $n = 1$? Calculate the corresponding wavelength of the emitted photon to support your answer.

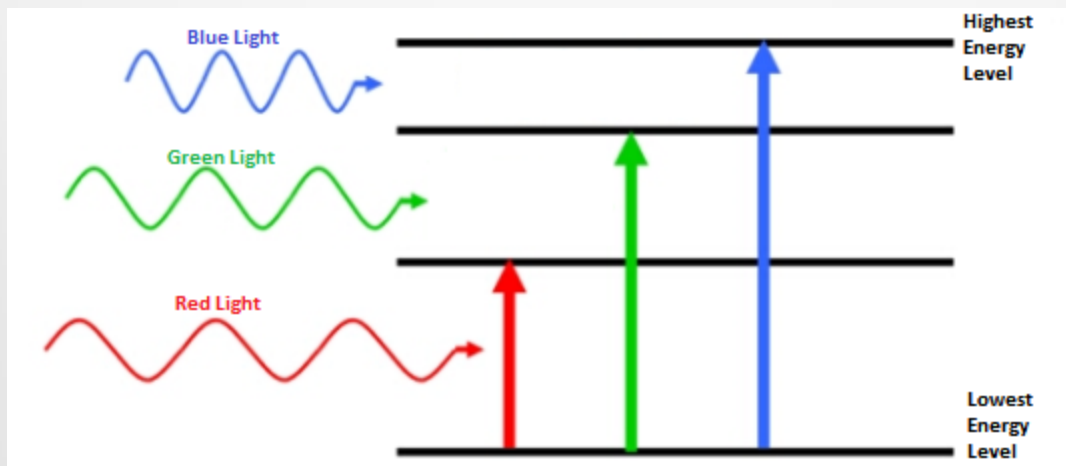
$$\Delta E = 10.2 \text{ eV}$$

Wavelength is 122 nm

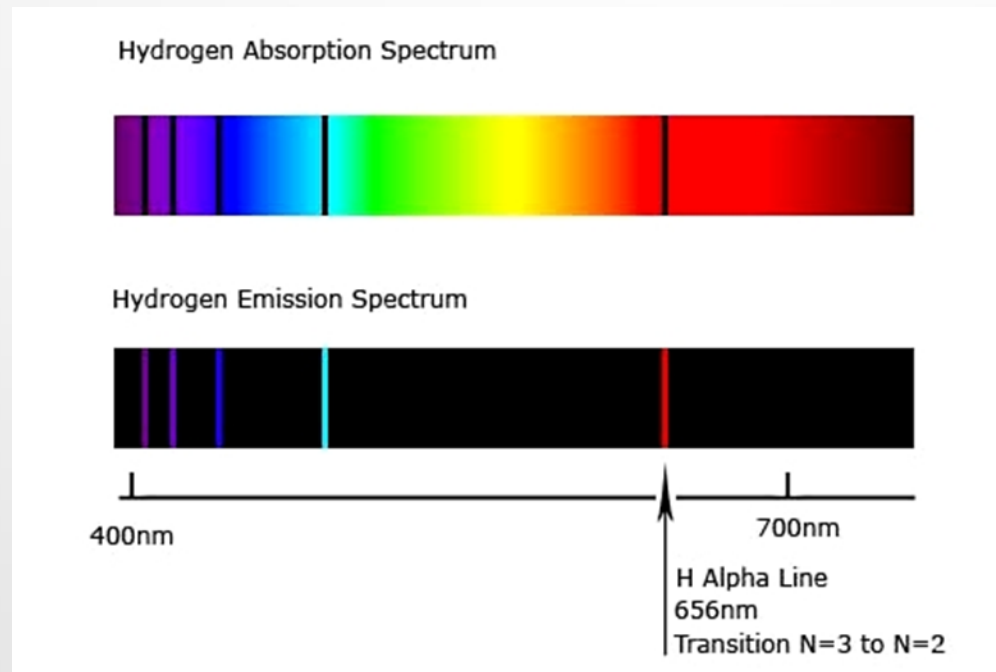
Photon is in UV range

Absorption

- When source of white light is passed through cold hydrogen gas the electron in the atoms can absorb a photon.
- The energy from the photon can be used to make the electron move up to a higher energy level.
- This only occurs if the photon has an energy that **exactly** matches the energy required to reach a higher energy level, or if the energy is greater than the ionisation energy (as the excess energy goes into the kinetic energy of the emitted electron).



- This means the photon is absorbed by an atom in the gas.
- If there are enough atoms in the gas, then the colour associated with this photon does not appear in the spectrum of light after it has passed through the gas.



Question: A hydrogen atom collides with an incident photon of frequency 3.42×10^3 THz. What is the kinetic energy of the ground state electron after it absorbs the photon?

Answer:

$$E = hf = (6.63 \times 10^{-34})(3420 \times 10^{12}) = 2.267 \times 10^{-18} J$$

$$E = 14.2 \text{ eV}$$

This does not correspond to a specific energy level transition, BUT it does ionise the electron as it exceeds 13.6 eV.

$$\Delta E = 14.2 - 13.6 = 0.572 \text{ eV} = 9.146 \times 10^{-20} J$$

Therefore the kinetic energy is 0.572 eV or $9.15 \times 10^{-20} J$